

## MAJOR CONTRIBUTORS TO THIS REPORT



**Abhinav Patil**,  
field applications  
manager - key  
accounts, Analog  
Devices India



**Ankan Mitra**,  
vice president,  
SMTA India Chapter



**Natarajan M.M.**,  
vice president - South  
Asia, Arrow Asia-  
Pacific



**Srinivasa  
Moorthy**,  
director, D4X  
Technologies  
Pvt Ltd



**Tim Patel**,  
technical marketing  
manager, Electronics  
Business Unit,  
Littelfuse Inc.

condition in wall-mount chargers will come directly from the wall into that charger. It is a very dangerous application if you do not have adequate circuit protection. The charger can explode, catch fire or put the other chargers bundled next to it on fire. To solve this potentially dangerous issue, a very small form factor axial fuse is placed inside wall-mount chargers. This is how changing form factors of end devices force small form factors for protection devices, too."

### Some golden rules to live by

If you have read up to this point, then you need not be told that circuit protection is quite a can of worms. You essentially have to weigh up risk, cost and consequences. In the end, though, you should not mind spending an extra ₹ 5000 on a circuit that costs ₹ 100,000, as opposed to a circuit that costs ₹ 10,000.

I talked to a host of designers and they all had their own views on what the golden rules are, which must be followed. But within the chaos of opinion, some guidelines did appear a bit more inline than others. Read on to find out if you follow these favourite guidelines of your peers.

According to Moorthy, "Always understand the standards that the product needs to comply with. This is the most significant input for a successful design. Take special care to note the environmental specs, in particular, humidity and altitude. Chances of ESD strikes are higher at low humidity, which can result in complete

circuit failure. Increasing chances of lightning at higher altitudes precipitates the need for additional lightning protection to be provided to mitigate the same. I would also suggest working closely with enclosure designers, especially when the enclosure is made of plastic."

Though nothing beats getting the required results in the lab where everything is hunky dory, there is a difference between theory and practical. If not installed properly, protection can end up damaging the thing it was installed to protect. As is now obvious, only the most thorough understanding of the aspects of circuit protection will be helpful in making your designs safe from all harm. At present rate, adoption of ESD, EMI and other phenomena into circuit-protection tools will soon be complete and early adoption of the miniaturised protection devices can only foster future devices with even longer lives. Long live machines!

### Little things to remember about little fuses

In choosing the right fuse for a specific application, you need to know the environment in which the fuse is being used. There are several factors used to select the proper fuse.

One is normal operating current. You need to gauge how much steady-state current is going through the system. A fuse has to operate without unintentionally bursting during the lifetime of the circuit. Fuses can weaken over their lifetimes by temperature cycling and by current that

causes heating. This causes incremental stressing of the fuse element and can cause brittleness of alloys. As a result, it can cause destruction of the fuse when it is not supposed to; this is called nuisance tripping.

The second factor is application voltage. Fuses are sensitive to changes in current, not voltage. It is not until the fuse element melts and arcing occurs that the circuit voltage and available power become an issue. A fuse may be used at any voltage that is less than its voltage rating without being detrimental to its fusing characteristics.

The third factor is ambient temperature, which can affect the fuse's reliability. Current-carrying-capacity tests of fuses are performed at 25°C and will be affected by changes in ambient temperature. The higher the ambient temperature, the hotter the fuse will operate and the shorter its life. Practical experience indicates, fuses at room temperature should last indefinitely, if operated at no more than 75 per cent of the catalogue fuse rating.

The next factor is maximum-fault or short-circuit current, also called interrupt rating or breaking capacity. A fuse must meet or exceed the maximum-fault current of the circuit. The interrupt rating is the maximum current that the fuse can safely interrupt at a rated voltage. During a fault or short-circuit condition, a fuse may be subjected to an instantaneous overload current many times greater than its normal operating current.

Pulses are another factor. Over its lifetime, a fuse can experience inrush currents that happen on startup. Electrical pulses produce thermal cycling and possible mechanical fatigue that could affect the life of the fuse. For some applications, startup pulses are normal and require the use of fuses that incorporate a thermal-delay design to enable these to survive normal startup pulses, while providing protection against prolonged overloads. It is important to define the startup pulse and then compare it to the fuse's time-current curve. ●